CHAPTER

CLINICAL APPLICATIONS OF BREAST BIOMECHANICS

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1 INTRODUCTION

Breast cancer is the most prevalent form of cancer in women (Boyle and Levin, 2008). Approximately 1.2 million new cases are diagnosed each year, and 400,000 women die from the disease each year (Veronesi et al., 2005). The USA National Cancer Institute estimates the lifetime risk for a woman to develop breast cancer at 12%. Early detection and treatment of breast cancer can help save lives. Despite promising progress in our understanding of breast cancer development and growth, clinicians are still faced with a number of challenges for improving patient outcomes.

1.1 DIAGNOSING BREAST CANCER

X-ray mammography is considered the gold standard, and the only feasible approach for early detection and diagnosis of breast cancer. Reasons for this include high specificity, high throughput (with scans typically taking 10 s), and a relatively low cost compared with other imaging modalities, such as magnetic resonance imaging (MRI), which can cost up to four times more than a mammogram (Kahán, 2011). During mammography, the breast is compressed between two plates, and X-rays are transmitted through the breast tissue to produce a two-dimensional (2D) projection image (Fig. 1A). Breast compressions of up to 7% are used to provide a more uniform reduced thickness of tissue, producing greater contrast between tumors and normal breast tissue, a key to successful detection of the disease. However, the difficulty in performing such large compressions in women with dense breasts limits the applicability of mammography, because of decreased image contrast, and therefore sensitivity (Lehman, 2006). In such cases, ultrasound imaging (USI), shown in Fig. 1B, can be used for investigating palpable abnormalities within dense tissue and for distinguishing between solid tumors and cysts (Malur et al., 2001). While being portable, real-time, inexpensive, and relatively easy to use, conventional ultrasound images are acquired in 2D and have relatively low spatial resolution compared with mammograms, making their analysis difficult. While tumors can be detected in both mammography and ultrasound images with a sensitivity of 80%–90%, their size may be significantly underestimated (Heusinger et al., 2005; Dummin et al., 2007; Mann et al., 2008a). Furthermore, mammography alone, or mammography combined with breast ultrasound, has been found to be insufficient for early diagnosis of breast cancer in women who have high lifetime risk of breast cancer (Kuhl et al., 2005).

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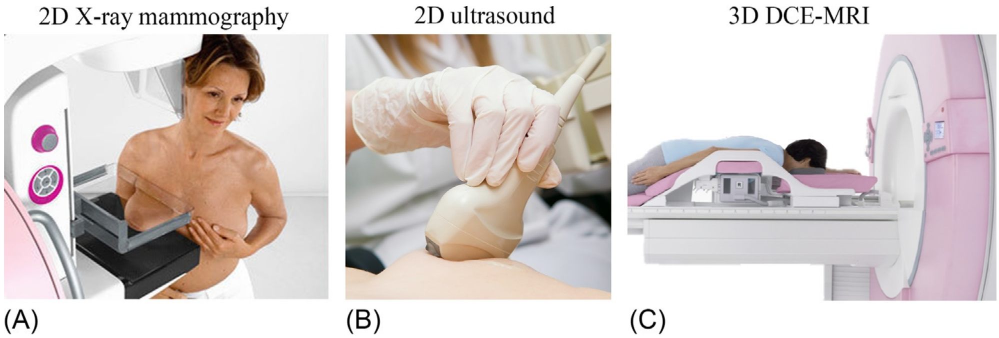

FIG. 1

Imaging modalities used for diagnosing breast cancer. (A) X-ray mammography involves compression of the patient's breast while standing. (B) Ultrasound imaging is performed while the patient is supine, and requires contact between the ultrasound probe and the skin surface. (C) MRI is typically performed while the patient is in the prone position, with arms positioned either at the side of the body or above the head. Part of the MR imaging procedure involves intravascular injection of a contrast agent, which is used to improve contrast between tumors and the remaining tissues.

Images from siemens.com/press.

3D MRI (Fig. 1C) provides a noninvasive means of diagnosing breast cancer, with a sensitivity close to 100% (Berg et al., 2004; Van Goethem et al., 2006). Tumor sizes of invasive breast cancers in MR images also correspond well with their size identified postexcision in pathology tests (Van Goethem et al., 2006; Mann et al., 2008a,b). However, despite its high sensitivity, randomized clinical trials have reported a low specificity in detecting cancers when only MRI is used during diagnosis. This typically results in a high false positive rate, leading to unnecessary surgical procedures (Turnbull et al., 2010; Peters et al., 2011).

Collocating suspicious tissue features between mammography, ultrasound, and MR images may help reduce the false positive rate associated with MR imaging. This also helps diagnose cancer with a significantly higher sensitivity, and at a more favorable stage in high risk women, compared with screening with only mammography and USI (Malur et al., 2001; Berg et al., 2004; Kuhl et al., 2005). However, the significantly different loading conditions that the breast experiences during these different imaging procedures (Fig. 1) makes collocation of regions of interest across the images a difficult task, and one of the main challenges hindering accurate diagnosis of breast cancer.

1.2 TREATING BREAST CANCER

Once the diagnosis of a suspected breast tumor has been confirmed by tissue biopsy, its position remains to be identified during treatment. Breast surgery to remove tumors is often the first step in treating early stage breast cancer. Two options exist for breast surgery: mastectomy, where the entire breast is removed; or breast conserving therapy (BCT) (also called *lumpectomy*), which aims to only remove the cancerous region, and a small amount of surrounding normal tissue called the margins (McLaughlin, 2013). Studies have shown that women with localized breast cancer are equally likely to survive their cancer when treated with mastectomy or BCT (McLaughlin, 2013). Because most early stage carcinomas are small, and large primary tumor sizes can be reduced by primary chemotherapy, BCT has become the most popular treatment, representing 75%–85% of all operations in most breast cancer centers (Veronesi et al., 2005).

Successful treatment of early-stage breast cancers using BCT is dependent on the accurate determination of clear surgical margins surrounding the tumors during surgery, to reduce the risk of local recurrence (Houssami et al., 2010, 2014; Chiappa et al., 2013). Complete excision of tumors depends on the ability of the surgeon to palpate the lesion in its entirety. This procedure is very demanding because: all imaging modalities acquire information with the patient in different positions to that assumed during surgery; and palpation of the tumor might be difficult, depending on the composition of the breast and the characteristics of the cancer, making intra-operative localization inaccurate (Turnbull et al., 2010). This is particularly important for invasive breast cancers, which have spread from the mammary glands into the surrounding breast tissue.

The difficulty in localizing tumor positions is significant, because the majority of studies in the literature report incomplete excision of tumors in 20%–40% of the patients who undergo BCT (Pleijhuis et al., 2009). In such cases, additional surgery is required to remove the remainder of the tumor. This results in risk of wound infection, poor cosmetic appearance, psychological distress, and increased treatment durations and health costs.

1.3 APPLYING BREAST BIOMECHANICS TO ADDRESS CLINICAL CHALLENGES

Biomechanical modeling of the breast has the potential to help clinicians address many of the challenges described in Sections 1.1 and 1.2, by simulating the deformations that the breast undergoes during different breast imaging procedures. The models can then be used to more easily map and colocate information between the various medical images to help diagnose breast cancer. Once a tumor has been identified, the models can also provide the ability to predict its location in the positions assumed during treatment procedures, such as surgery. Such applications typically require biomechanical models to be customized to an individual, and therefore the subject-specific geometry of the individual breast tissues need to be modeled. Furthermore, the mechanical properties of these tissues need to be identified in order to give reliable predictions of their mechanical behavior.

This review describes the state of the art in the field of breast biomechanics modeling. It begins by briefly outlining the anatomy of the breast in Section 2. This is followed in Section 3 by an overview of how this anatomy is represented using individual-specific finite element (FE) models of the breast. The theoretical aspects of simulating breast deformation under different clinical procedures is then presented in Section 4, followed in Section 5 by a description of how the constitutive behavior of the different breast tissues has been defined through the assignment of individual-specific tissue mechanical properties. Section 6 provides examples of how breast models can be used in different applications. Lastly, Section 7 describes some of the technical and practical challenges that need to be addressed before these models can be implemented in clinical workflows.

2 ANATOMY OF THE BREAST

The breasts of an adult woman are tear-shaped glands situated on the chest (Fig. 2), and are responsible for lactation (Drake et al., 2014). They extend between rib II and VI, and primarily sit on the pectoralis major muscle, which is responsible for movement of the shoulder joint. The pectoralis major originates from the medial surface of the clavicle and the anterior surface of the sternum, and inserts into the lateral lip of the bicipital groove of the humerus. At the axillary margins, the breasts sit on the pectoralis minor, serratus major, external oblique muscles, and the fascia of the thorax.

The internal structures of the breast are contained within the superficial and deep (pectoral) fascia shown in Fig. 2. Within these two layers lie the secretary glands (lobules), which are the functional units that produce milk. The lobules connect through a network of lactiferous ducts to the nipple. The ducts and lobules are surrounded by connective tissue, composed of mainly fibrous and adipose

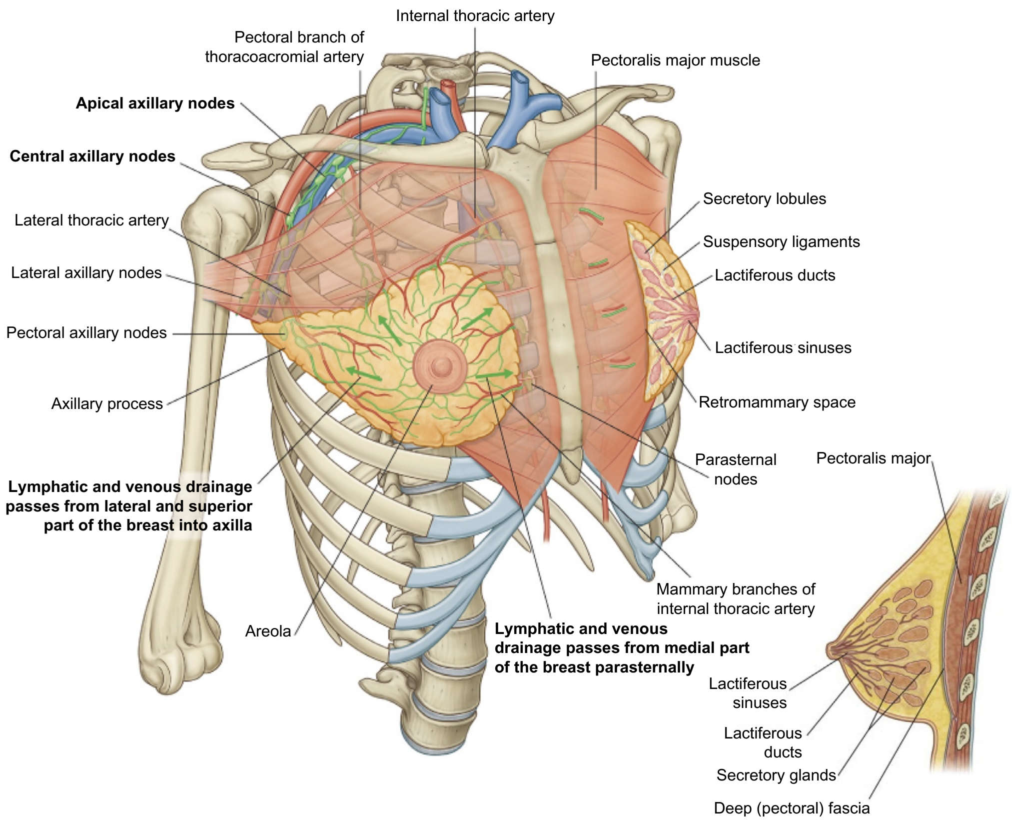

FIG. 2

Anatomy of the breast.

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tissue, along with blood vessels, lymphatic vessels, and nerves. The proportion of fibroglandular to adipose tissue within the breast varies between individuals, and is dependent on age, with the fibroglandular to adipose ratio decreasing with age (Kopans, 2006).

The internal tissues of the breast are supported by fibrous suspensory ligaments known as Cooper's ligaments, which extend between the superficial and deep fascia. These ligaments provide suspensory support, and allow the breast to maintain their normal shape (Cooper, 1840; Riggio et al., 2000). A layer of loose connective tissue, named the retro-mammary space, separates the breast from the deep fascia, enabling a relative sliding motion between the functional tissues of the breast tissue and the underlying pectoralis major muscle. A detailed review of the anatomy of the breast fascial system is available in Riggio et al. (2000).

2.1 3D PATIENT-SPECIFIC MODELING OF BREAST ANATOMY

Patient-specific models of the breast have been created predominantly from MR images acquired in the prone position. This provides 3D information regarding the boundaries (skin and air, breast tissue and muscle, muscle and rib), as well as the internal structure of the breast. While less common, computed tomography (CT) images in the prone or supine positions have also been used for model construction in breast augmentation applications (Roose et al., 2005; del Palomar et al., 2008).

Different approaches have been used to segment the boundaries of the breast from MRI or CT images. Some methods use intensity thresholding with morphological operations (Twellmann et al., 2005), or edge-based constraints (such as sign of gradients (Yao et al., 2009), a Hessian filter (Wang et al., 2012), or a tuneable Gabor filter (Milenković et al., 2015)) for breast segmentation. Others use probabilistic atlases (Ortiz and Martel, 2012; Gubern-Mérida et al., 2015), or supervoxel classification techniques (Baluwala et al., 2015) to segment the boundaries of the breast. Once the boundaries of the breast have been segmented, a number of meshing algorithms have been employed to construct anatomically realistic FE models for a specific individual. These include Delaunay triangulation, marching cube algorithms (Tanner et al., 2006a,b; del Palomar et al., 2008), and nonlinear geometric fitting (Nielsen, 1987; Fernandez et al., 2003; Rajagopal et al., 2008a). Tetrahedral (Roose et al., 2005; del Palomar et al., 2008; Tanner et al., 2009) and hexahedral (Ruiter et al., 2006) elements are the most widely used element types when creating FE meshes, and are usually interpolated using standard Lagrange-based interpolation schemes, such as linear or quadratic Lagrange shape functions. Higher order C^1 continuous Hermite-based interpolation schemes (Bradley et al., 1997) have also been used to represent the geometry of the breast, as they provide gradient (and hence strain) continuity, and faster rates of solution convergence (Strang and Fix, 2008). Fig. 3 shows an example of the use of cubic Hermite hexahedral elements and nonlinear geometric fitting for creating an individual-specific FE mesh of the breast from segmented MRI data in the prone orientation.

Fat, fibroglandular, pectoral muscle, and tumor tissues within the breast can be segmented from medical images in a number of ways, such as manual segmentation (Azar et al., 2000), standard thresholding (Samani et al., 2001; Pathmanathan et al., 2008), or fuzzy C-means (FCM) based segmentation (Shih et al., 2010). Segmentation algorithms in a number of software packages have also been used to segment the different tissues, including ANALYZE (Biomedical Imaging Resource, Mayo Foundation, Rochester, MN, USA) (Tanner et al., 2001, 2009) and Scion Image (Scion Corp., Maryland USA) (Azar et al., 2000). Fig. 3C shows an example of a pectoral muscle mesh within a breast model, overlaid with a segmentation of the fibroglandular tissue region that was generated using a standard thresholding

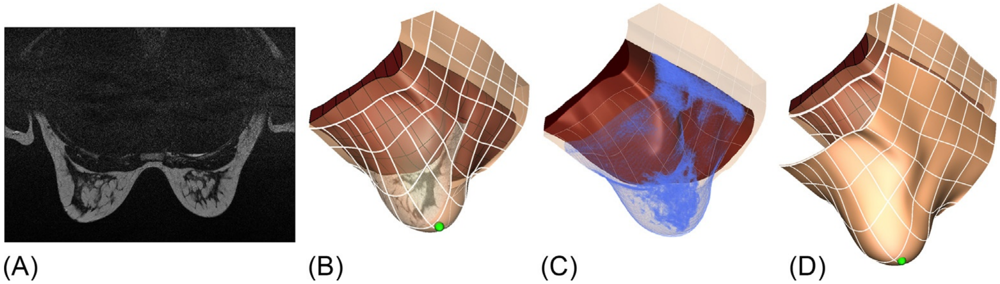

FIG. 3

Construction of an individual-specific FE mesh of the breast (composed of pectoral muscle tissue and breast tissue), created by fitting cubic Hermite hexahedral elements to segmented boundaries from breast MRI, is shown in (A,B). The fibroglandular tissue, segmented using a standard thresholding approach, is shown in (C). A skin mesh of constant thickness, extracted from the skin surface boundary of the fitted breast FE mesh is shown in (D) (the skin mesh has been offset from the breast surface for visualization purposes).

technique. The skin on the breast can be modeled using 2D or 3D elements that are coupled to the surface of the 3D breast mesh (Ruiter et al., 2006; del Palomar et al., 2008; Tanner et al., 2009; Solves-Llorens et al., 2012). In general, the thickness of the breast skin cannot be determined directly from MR images, due to their relatively low spatial resolution, compared to, for example, CT images. Therefore, the skin thickness is often prescribed based on measurements reported in the literature. For example, (Huang et al., 2008) measured from CT images breast skin thickness of 1.45 mm $\pm$ 0.30 mm, with a range of 0.9–2.3 mm in 51 individuals. Based on these measurements, Fig. 3D shows an example of a skin layer of constant thickness of 1.45 mm that has been added to the surface of a breast model.

Due to the difficulties in identifying and segmenting the geometric arrangement of Cooper's ligaments within the breast, Cooper's ligaments have not yet been explicitly modeled. However, their mechanical influence has been investigated indirectly, through the use of anisotropic tissue descriptions (discussed further in Section 5). The remaining structures and vessels (such as blood vessels and lymphatics) are usually not modeled discretely because their contribution to the overall mechanical response is assumed to be minor compared with the tissues described above. A survey of the different breast tissues that have been modeled in previous studies is included in Table 1.

3 SIMULATING BREAST BIOMECHANICS

This section provides an overview of the theoretical and computational techniques that have been used to model the biomechanics of the breast, subject to the various loading conditions applied during clinical procedures.

Patient-specific breast deformations are often simulated using continuum mechanics and the FE method. Each of the breast tissues is typically constrained to be incompressible due to the large water content (Fung, 1993). In the cases of relatively small deformations, small strain mass-tensor approximations have been used, such as in the case of aiding registration of temporal mammograms acquired in the same position for temporal tracking of the progression of tumors (Roose et al., 2006a, 2008).

Table 1 A Survey of Recent Breast Biomechanical Models

Lead Author	Modeling Theories	Imaging Modalities/Degree of Deformation	Parameters/ Anatomy	Boundary Conditions[a]	Performance[b]
Colocating tumors between different diagnostic imaging modalities					
Ruiter et al. (2004, 2006)	Nonlinear	MR mammography and X-ray mammography/21% compression	Literature/ homogeneous skin	Prescribe plate displacement	4.3 mm (X-ray mammography) and 3.9 mm (MR-mammography) landmarks error
Roose et al. (2006a, 2008)	Linear, mass-tensor model	Temporal prone-MR/n/a	Literature/ homogeneous	Prescribe skin surface displacement	70%–80% decrease in SSD of image intensities between follow-up MR images
Sarkar et al. (2007)	Linear	X-ray mammography (CC, MLO)/n/a	Literature/ homogeneous	Prescribe plate displacement	2.3 mm mean landmark error
Shih et al. (2010)	Nonlinear, frictional contact	MR-mammography/ 20%–60% compression	Literature/ adipose, fibro	–	Not reported
Han et al. (2010)	Nonlinear	MR-mammography/ 30% compression	In vivo/ adipose fibro, muscle	Prescribe plate displacement	<4.2 mm landmark error
Lee et al. (2010a)	Nonlinear, frictionless contact	MR-mammography/ up to 38% compression	Literature/ homogeneous	Rib-muscle tied contact	4.4 mm RMS registration error
Reynolds et al. (2011)	Nonlinear, frictionless contact	X-ray mammography, prone-MR/up to 75% compression	Literature/ homogeneous	Rib-muscle tied contact	Jaccard coefficient of 14%–75% for tumors postcompression
Tanner et al. (2011)	Linear	MR-mammography/ up to 59% compression	In vivo/fibro, muscle, skin, tumor	Prescribe surface displacement	4.1 mm landmark error
Tracking tumors for surgical assistance					
Rajagopal et al. (2007b, 2008b)	Nonlinear	MR/prone-to-supine	In vivo/ homogenous	–	7.7–8.4 mm skin surface RMS error
Carter et al. (2008, 2012)	Linear and nonlinear	MR/supine-to-prone	In vivo/ adipose, fibro	Prescribe skin surface displacement	6.8–10.3 mm landmark registration error between prone and supine images

Continued

Table 1 A Survey of Recent Breast Biomechanical Models—cont'd

Lead Author	Modeling Theories	Imaging Modalities/Degree of Deformation	Parameters/ Anatomy	Boundary Conditions[a]	Performance[b]
Babarenda Gamage et al. (2012)	Nonlinear	MR/prone-to-supine	In vivo/ adipose, muscle	–	5.0 mm skin surface RMS error
Eiben et al. (2013)	Nonlinear	MR/prone-to-supine, supine-to-prone	In vivo/ adipose, fibro, muscle	Prescribe muscle surface displacement	5.3–6.8 mm landmark registration error between prone and supine images
Han et al. (2011, 2012, 2014)	Nonlinear, frictional contact	MR/prone-to-supine	In vivo/ adipose, fibro, muscle, skin	–	4.4–13.3 mm landmark registration error between prone and supine images
Predicting outcomes of breast augmentation procedures					
Roose et al. (2005)	Mass-spring model (non-FE)	CT, stereoscopy/ pre-post implant	Literature/ adipose fibro	n/a	8.5 mm maximum error between pre-post implant skin surface
Roose et al. (2006b)	Linear, mass-spring model	Stereoscopy/pre-post implant	Literature/ homogeneous	Prescribed implant displacement	4 mm mean error between pre-post implant skin surface
del Palomar et al. (2008)	Nonlinear	CT, stereoscopy/ supine-to-standing	In vivo/ adipose, fibro	–	2.4 mm skin surface error in standing position tumors

n/a, *not available;* SSD, *sum of squared distances;* RMS, *root mean square.*
[a]*All models assumed fixed chest wall (rib-muscle or muscle breast tissue) unless otherwise stated.*
[b]*Performance measures of the models were obtained by comparing model predicted locations/features/regions with independently acquired images or skin surface reconstructions.*

For larger deformations, linear elasticity or pseudo-nonlinear elasticity have been used as a trade-off between computational efficiency versus the biophysical accuracy provided by the nonlinear finite strain elasticity formulation. When modeling large deformations, excessive mesh distortion can be problematic. To address this issue, researchers have employed remeshing, or mesh moving, algorithms. These aim to generate new meshes that are less distorted, and to remap previously computed solutions onto these new meshes (Stein et al., 2004). Such studies have been performed in the context of breast modeling where remeshing was shown to provide good numerical agreement between remeshed linear elasticity and nonlinear finite strain solutions for up to 30% compression of a cube

(Tanner et al., 2006a,b). However, with only relatively small compressions being applied in these studies, it is unclear whether these methods can provide accurate solutions for clinical compression (up to 75% compression in extreme cases), without introducing significant errors due to small strain approximations. Pseudo-nonlinear elasticity, which assumes the linear elastic modulus (or Young's modulus) is a function of strain, has been used to model breast compression for biopsy procedures (Azar et al., 2001, 2002; Yin et al., 2004), and to simulate deformation due to gravity in the standing position for modeling static and dynamic thermography of the breast (Jiang et al., 2011). Numerical studies have highlighted differences between linear, pseudo-nonlinear, and fully nonlinear continuum mechanics formulations under different gravity loading conditions, and concluded that both linear and pseudo-nonlinear elasticity models were poor approximations to finite deformation models (Whiteley et al., 2007). A number of studies have thus employed fully nonlinear large deformation theory to model deformations due to gravity, and for large degrees of compression (see Table 1).

3.1 SIMULATING BREAST DEFORMATION UNDER GRAVITY LOADING

Simulating deformation of the breast under gravity loading is important for many clinical applications, such as tracking features of interest within the breast between the prone and supine positions. This is achieved by changing the orientation of the model with respect to the direction of gravity. The nonlinear large deformation theory used to simulate breast biomechanics during such reorientation typically assumes that the breast is initially in a stress-free reference configuration. However, it is important to note that the breast is generally under the influence of gravity during an imaging procedure. Therefore, errors may arise in model predictions if, for example, the prone configuration is used as the stress-free reference configuration when predicting, say, the supine gravity-loaded configuration. This is because the prone configuration is not the stress-free reference state of the breast. Moreover, simply applying the effect of gravity in the opposite direction does not yield the correct stress-free state.

A numerically valid procedure for determining the stress-free reference state using a displacement/pressure-based Lagrangian finite element formulation was described by Rajagopal et al. (2005), Rajagopal et al. (2006), and Rajagopal et al. (2007a). These studies included validation experiments on physical phantoms to quantify the accuracy of this technique in practice. Similar studies on phantoms were performed by Carter et al. (2009). More recently, Vavourakis et al. (2015) described a displacement/pressure-based approach for determining the stress-free reference state, similar to that described in Rajagopal et al. (2007a), but based on a Eulerian formulation. A limitation of these approaches is that the accuracy of the estimated stress-free reference state is largely influenced by factors such as the boundary conditions and assumed material parameters specified in the model, as well as the accuracy to which the geometry of the gravity loaded tissues has been quantified. In light of these limitations, studies have been performed to assess how well the stress-free reference state of the breast can be numerically approximated through neutral buoyancy experiments (Rajagopal et al., 2007b, 2008a,b; Carter et al., 2009), where a participant's breasts were immersed in water during MR imaging in the prone position. In these experiments, the density of the breast was assumed to be similar to that of water. Fig. 4A illustrates the procedure used to compare an estimate of the stress-free reference state, determined using the algorithm described in Rajagopal et al. (2007a), with neutral buoyancy experiments. While performing such neutral buoyancy experiments with MR imaging is impractical in the clinical setting, such experiments provide a method for validating the numerical approaches for determining the stress-free reference state of the breast.

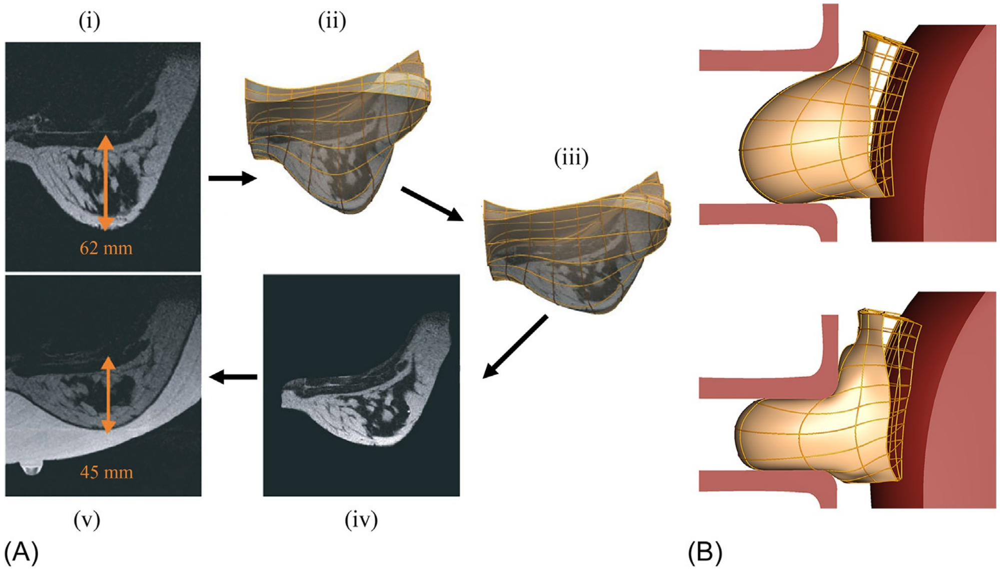

FIG. 4

(A) An illustration of the method used for estimating the stress-free reference state of the breast described in Rajagopal (2008b). (i) MR image of the breast of a volunteer in the prone gravity-loaded state. (ii) Prone MR image embedded in a finite element model that was fitted to the prone 3D MR dataset. (iii) Predicted stress-free reference sate of the breast where the prone image has been warped to generate (iv) a predicted MR image in the stress-free reference configuration, which can be compared against (v) an acquired MR image of the neutrally buoyant breast immersed in water (indicated by the large bright region in the image). The tissue displacement between this neutral buoyancy image and the prone gravity-loaded image in (i) can be estimated by comparing the distance between the nipple and pectoral muscle surface. (B) An example of CC mammographic compression using frictionless contact mechanics and finite strain theory (Chung et al., 2008a).

3.2 SIMULATING MAMMOGRAPHIC COMPRESSION

During X-ray mammographic imaging, the breast is compressed between two plates. This is typically performed in two directions: the craniocaudal (CC) view involves compression in the head-to-toe direction; and mediolateral oblique (MLO) view involves compression in the shoulder-to-opposite-hip direction. Clinical X-ray mammography usually involves compressions of between 20% and 75% of the unloaded breast thickness, depending on the size and density of breast tissues. While the distributions of boundary loads across the surface of the breast during compression procedures are not routinely measured, the separation of the compression plates is typically recorded (although in practice, the plates may not be parallel, and the position of the plates relative to the patient is often only qualitatively characterized).

Breast compression has previously been modeled using two main methods. The first involves the direct application of displacements to nodes on the surface of a FE model to mimic the mechanical

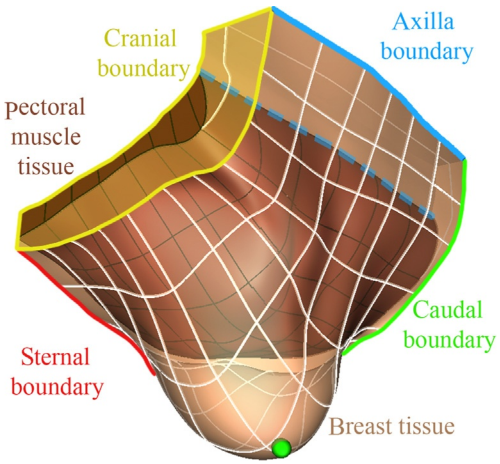

FIG. 5

Regions of the breast where boundary conditions are typically specified.

effects of the plates. The second, more physically realistic, approach to simulating breast compression involves applying contact mechanics constraints between the skin and the plates. Readers are referred to Laursen (2003) for more information regarding the theoretical formulation of the contact mechanics constraints for small and finite strain continuum mechanics problems. The theoretical aspects regarding the application of contact mechanics to breast compression are described in Chung et al. (2007), Chung (2008a), Chung et al. (2008b). Controlled studies using phantoms have been performed, to validate the applied theoretical and numerical methods, on simple geometries (Samani et al., 2001; Qiu, 2003; Chung, 2008; Lee et al., 2010a; Lee et al., 2010b). The main purpose of these studies was to determine the accuracy with which the numerical methods for contact mechanics can be applied under idealized conditions.

Patient-specific clinical compression studies have been performed by a number of authors (see Table 1). In previous studies, the contact constraints between the breast and the plates were simulated by assuming either frictionless interactions (Chung, 2008; Reynolds et al., 2011; Han et al., 2012; Hopp et al., 2012; Lee et al., 2013; Mertzanidou et al., 2014), or frictional interactions (Alonzo-Proulx et al., 2010; Shih et al., 2010; Hopp et al., 2013). The largest compression simulations performed to date involved up to 75% compression on clinical datasets (Reynolds et al., 2011). Fig. 4B provides an example of a patient-specific CC compression using frictionless contact mechanics constraints and finite strain theory.

3.3 BOUNDARY CONDITIONS

The majority of breast biomechanical models described in the literature only consider one side of the breast. Fixed boundary conditions are usually applied in the region corresponding to the sternal boundary, and on the cranial and caudal boundaries of the model (see Fig. 5). Boundary conditions are also typically applied to restrict the motion of the posterior surface of the breast and axilla boundary during studies involving gravity or compression loading. In most cases, these surfaces are assumed to be fixed (e.g. see Babarenda Gamage et al., 2012), or it is assumed that the breast tissue slides over the rib-pectoral muscle interface, which shifts the axilla boundary further towards the back of the body (e.g. see Lee et al., 2013).

4 MECHANICAL PROPERTIES OF BREAST TISSUES

Robust identification of the parameters describing the mechanical behavior of the different breast tissues is important for providing accurate predictions of the overall organ behavior. This is difficult due to the heterogeneous distribution of the mechanical properties of different breast tissues (Krouskop et al., 1998; Wellman et al., 1999). There is also significant variation in these mechanical properties across the population, due to factors such as age, physiological conditions (for example, the menstrual cycle, pregnancy, and the onset of menopause), and the pathophysiological state of a given individual (Krouskop et al., 1998; Wellman et al., 1999; Sinkus et al., 2002).

Previous studies have estimated mechanical properties of the breast from ex vivo samples of tissue under compression (less than 15% (Wellman et al., 1999), less than 25% (Krouskop et al., 1998)), o indentation (Samani and Plewes, 2004; Kerdok et al., 2005; Samani et al., 2007; Samani and Plewes, 2007; O'Hagan and Samani, 2009,) loading. The mechanical properties of tumors have also been identified ex vivo (Krouskop et al., 2003; Samani et al., 2003; Samani and Plewes, 2007; O'Hagan and Samani, 2008; O'Hagan and Samani, 2009; Wessel et al., 2010). These studies involved fitting parameters to different phenomenological hyperelastic constitutive relations. These include linear, piecewise linear, exponential, Yeoh, Ogden, and Arruda-Boyce constitutive relations (Liu et al., 2004). Similar methods have also been used to identify the mechanical properties of tumors in excised samples of breast tissue (Krouskop et al., 2003; Samani et al., 2003; Samani and Plewes, 2007; O'Hagan and Samani, 2008; O'Hagan and Samani, 2009). Examples of ex vivo parameters previously identified in the literature are presented in Table 1, which illustrates the wide variability of reported parameters.

Although ex vivo experiments provide useful information about the orders of magnitudes of the mechanical properties, there are numerous drawbacks in using such parameters in patient-specific biomechanical models of the breast. One of the main issues is the significant variation of these parameters across the population, due in part to the range of different experimental conditions under which the mechanical tests were performed. Moreover, microstructural and physiological changes of the tissues occur over time due to tissue deterioration (Kerdok et al., 2006). Another drawback is the detachment of tissue from its surroundings during testing, which may contribute to the observed mechanical behavior, and would therefore influence the identification of parameters (Berardesca et al., 1995; Elsner et al., 2002; Lim et al., 2008). Due to these drawbacks, constitutive parameters have often been identified using in vivo measurements, with patient-specific biomechanical FE models of the breast, in order to obtain personalized estimates of the parameters. This has been achieved by applying a well-defined loading condition to an individual's breasts. Patient-specific models were constructed to simulate this loading condition. The tissue parameters describing the mechanical behavior of, for example, fat, fibroglandular, and muscle tissues were optimized to provide a match to the observed deformation (e.g. of the skin surface). Such investigations have been performed using MR-mammography, which involves compressing the breast during MR imaging in the prone position, or from gravity induced changes in breast shape during repositioning, for example, from the prone to supine positions. These studies have mainly used an isotropic neo-Hookean strain energy density function (SEDF) to describe the mechanical behavior of the breast tissues (Rajagopal et al., 2007b; del Palomar et al., 2008; Rajagopal et al., 2008a; Tanner et al., 2011; Carter et al., 2012; Eiben et al., 2013; Lee et al., 2013; Han et al., 2014). The incompressible form of this SEDF is defined in Eq. (1).

$$W = \frac{\mu}{2}(I_1 - 3) \tag{1}$$

In this relation, μ represents the small strain shear modulus of the tissue and I_1 is the first invariant of the right Cauchy-Green deformation tensor. A different shear modulus is often identified for each of the tissues that are included in the FE model.

Some studies have also investigated the use of anisotropic constitutive relations, particularity during MR-mammography, in an attempt to incorporate the effects of Cooper's ligaments into the breast models (Tanner et al., 2009; Han et al., 2012). For example, Han et al. (2012) used the SEDF shown in Eq. (2). This SEDF includes the compressible neo-Hookean isotropic component (W_{iso}) and the transversely isotropic component (W_{trans}) defined in Eq. (3).

$$W = W_{iso}(I_1, I_3) + W_{trans}(I_4) \tag{2}$$

$$W_{iso} = \frac{\mu}{2}(I_1 - 3) + \frac{k}{2}(J - 3), \quad W_{trans} = \frac{\eta}{2}(I_4 - 1)^2 \tag{3}$$

In this relation, k is the bulk modulus, η defines the modulus in the preferred direction, and I_3 and I_4 are the 3rd and 4th invariants of the right Cauchy-Green deformation tensor (with the latter describing the square of the stretch in the preferred direction). The Young's modulus (E) and Poisson's ratio (v) can be determined from μ and k using the following relations:

$$E = 2\mu(1 + v), \quad v = \frac{3k - E}{6k}, \tag{4}$$

Tables 2 and 3 present examples of parameters identified using these SEDFs. It is worth noting that parameter values identified in vivo using optimization-based approaches and FE models are, in general, dependent on the number of tissues modeled, and the accuracy to which boundary conditions have been applied. For example, if skin is not included in the model, the apparent stiffness of the remaining tissues would increase to compensate for the lack of a skin. Care should therefore be taken when comparing parameters identified from different models.

Table 2 A Summary of Previously Identified Isotropic Mechanical Properties of the Breast, Illustrating the Wide Variations of Parameters Identified Both ex vivo and In Vivo

Lead Author	Mode of Deformation	Constitutive Relation	Young's Modulus (kPa)				
			Adipose	Fibroglandular	Muscle	Skin	Tumor[a]
Krouskop et al. (1998)	Ex vivo compression	Linear elastic	20 ± 8 at 20% compression	48 ± 15 at 20% compression	n/a	n/a	290 ± 67
Wellman et al. (1999)	Ex vivo punch indentation	Exponential[b]	17.4 ± 7.9 at 15% strain	271.8 ± 167.7 at 15% strain	n/a	n/a	2162.1
Samani et al. (2003, 2007), O'Hagan and Samani (2008)	Ex vivo compression	Linear elastic	3.25 ± 0.9 at 5% compression	3.24 ± 0.61 at 5% compression	n/a	n/a	16.38 ± 1.55
del Palomar et al. (2008)	In vivo gravity loading	Neo-Hookean hyperelastic	18	52	n/a	n/a	n/a
Rajagopal et al. (2008b)	In vivo gravity loading	Neo-Hookean hyperelastic	0.48–0.90				n/a
Han et al. (2014)	In vivo gravity loading	Neo-Hookean hyperelastic	4.68 ± 3.07	5.91 ± 2.85	7.26 ± 2.52	8.31 ± 0.98	n/a

[a]*Ductal carcinoma in situ (DCIS).*
[b]*Wellman et al. (1999) derived average Young's moduli from exponential fits to their experimental data. These values are shown in the table.*

Table 3 Anisotropic Mechanical Properties of the Breast Identified by Han et al. (2012) From Mammographic Plate Compression Experiments on Five Subjects

Tissue Type	E/E_a	v	η/E
Adipose	1.00	0.4836 ± 0.0201	9.18 ± 8.72
Fibroglandular	15.51 ± 28.22	0.4936 ± 0.0098	15.09 ± 17.19
Muscle	42.74 ± 38.21	0.4761 ± 0.0168	16.49 ± 25.88
Tumor	98.60	0.4988	40.25

Kinematic boundary conditions were only applied during the FE modeling, therefore, only the relative values of the parameters could be identified. The adipose tissue (E_a) *was used as the reference material. Tumors were only present in one subject.*

5 APPLICATIONS OF PATIENT-SPECIFIC BREAST BIOMECHANICS

Several clinical applications of patient-specific biomechanical models of the breast are discussed in this section.

5.1 COLOCATING TUMORS BETWEEN DIFFERENT DIAGNOSTIC IMAGING MODALITIES

Medical image registration techniques provide a means for colocating regions of interest within the breast. Previous work has mainly focused on using nonrigid image registration algorithms, which involve applying a transformation from one image in order to best match image intensity values in another image. These transformations are typically determined using nonlinear optimization (Rueckert et al., 1999; Wirth, 2000; Dawant, 2002; Tanner et al., 2002; Behrenbruch et al., 2004; Sivaramakrishna, 2005; Guo et al., 2006). However, such methods permit nonphysical solutions, as they do not relate the registered motion to the underlying mechanical behavior of the tissues (which are governed by the fundamental laws of mechanics). To partially address these shortcomings, volume preserving algorithms have been applied to reduce tissue distortion during registration by applying more physically realistic constraints. These algorithms have been applied, for example, to register pre- and postcontrast MR images of the breast (Tanner et al., 2002; Rohlfing et al., 2003a; Rohlfing et al., 2003b). However, standard intensity-based techniques can fail if the observed deformation is too large (e.g. for large mammographic compressions, or for prone-to-supine repositioning), because the optimization objective function is typically highly nonlinear. Therefore, many studies have investigated using biomechanical models to provide physically realistic initial estimates of tissue motion in order to improve results of existing image registration algorithms (Hipwell et al., 2016).

Previous work in this area includes the use biomechanical models for:

- temporal registration of follow-up MRI images (Roose et al., 2006a; Roose et al., 2008);
- registration of mammogram images with MR images (Azar et al., 2001; Samani et al., 2001; Qiu, 2003; Qiu et al., 2004; Yin et al., 2004; Ruiter et al., 2006; Tanner et al., 2006a, 2006b; Hipwell et al., 2007; Kellner et al., 2007; Sarkar et al., 2007; Chung, 2008; Tanner et al., 2009; Han et al., 2010; Shih et al., 2010; Tanner et al., 2010);
- temporal registration of mammograms (Qiu et al., 2008; Qiu, 2009); and
- registration of X-ray mammograms and 3D ultrasound images (Hopp et al., 2015).

A survey of the performance of these breast models is provided in Table 1.

As an example, we highlight how a biomechanical model of the breast can be used to colocate breast lesions between X-ray and MR images, to help in the diagnosis of breast cancer. This application is of particular importance for women with high lifetime risk of breast cancer, where, as described in Section 1.1, the acquisition of MR and X-ray images are beneficial for strengthening diagnosis of the disease. However, X-ray imaging requires the breast to be significantly compressed while a patient is upright, and only produces 2D projection images. Diagnostic MR imaging, on the other hand, is generally performed in the prone position, with the breast hanging pendulously, as shown in Fig. 1C. This, coupled with the differences in contrast between the imaging modalities, can make it difficult for clinicians to colocate lesions of interest between these two types of images.

The use of physics-based biomechanical models can help in this respect. For example, in Lee et al. (2013) a 3D breast model was created from a set of prone MR images. The model was used to simulate the compression applied during the mammography procedure, and the predicted displacement field was used to warp the prone MRI to generate model-predicted MR images of the compressed breast in both the CC and MLO configurations. A perspective ray-casting algorithm (Hipwell et al., 2007; Pereira et al., 2010; Lee, 2011) was then used to generate a 2D pseudo-X-ray projection (Lee, 2011; Mertzanidou et al., 2014) from the warped 3D MRI. The pseudo X-ray images generated from the CC and MLO compressions were then aligned with real CC and MLO X-ray images, using a rigid registration algorithm, as shown in Fig. 6A. This procedure allows suspicious lesions identified in the prone MR images to be mapped onto the CC and MLO X-ray images. The accuracy of this technique is heavily influenced by the relative positions of the compression plates and the breast. However, this is complicated by the fact that such information is generally not recorded in clinical practice. In order to improve agreement between the pseudo and real X-ray images, the position and orientation of the plates can be iteratively updated, and the above procedure repeated until the best match between the pseudo and real X-ray images is achieved (Hopp et al., 2013; Lee et al., 2013; Mertzanidou et al., 2014). Solves-Llorens et al. (2014) followed a similar approach to Lee et al. (2013), with the 2D rigid registration between the pseudo mammogram and the actual mammogram replaced with a 2D nonrigid registration algorithm. However, since the nonrigid registration is only performed in 2D, the procedure may result in nonphysical registration results, especially if simplified models are used for simulating breast biomechanics.

5.2 TRACKING TUMORS FOR SURGICAL ASSISTANCE

Once suspicious lesions have been identified, their positions need to be tracked during various treatment procedures, such as BCT. As described in Section 1.2, such treatment procedures are usually performed in a different position to that used during diagnostic imaging, making it difficult to precisely identify the locations of the lesions due to the different loading conditions applied to the breast. Biomechanical models of the breast can help in this respect, as they can be used to track tumors, for example, from their location identified in a prone MR image to the supine position, in which several surgical procedures are performed.

Previous studies have investigated the use of preoperative supine MRI for helping plan surgical interventions (see Table 1). In such cases, biomechanical models can be used to register regions of interest between the diagnostic prone MRI and the preoperative supine MRI, as illustrated in

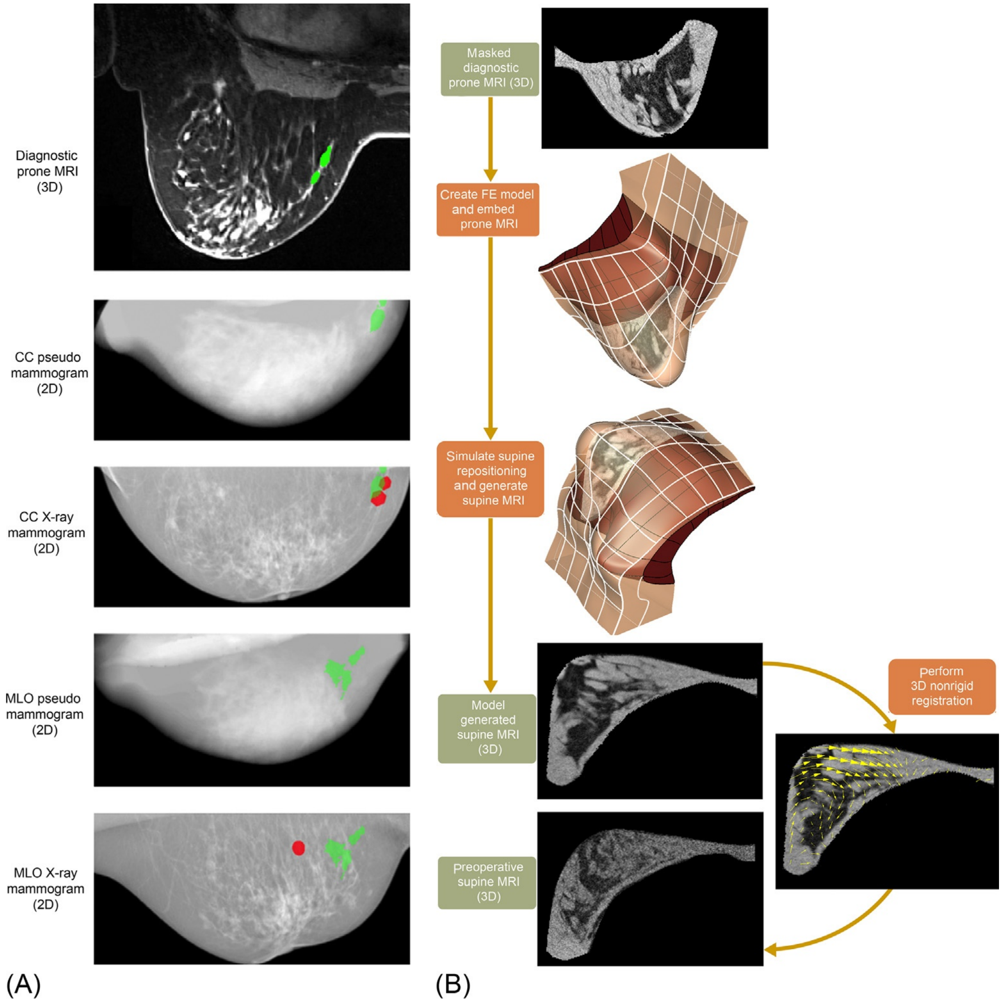

FIG. 6

(A) A comparison between pseudo mammograms generated from diagnostic prone MR images of a patient using the modeling framework described by Lee et al. (2013), and X-ray mammograms (see text for details). Green regions represent expert segmentation of lesions in prone diagnostic breast MR images *(top)* that were tracked to the pseudo-mammograms using the modeling framework. Red regions represent expert segmentations of the lesions in X-ray mammograms. (B) An example of how a biomechanical model can be used to simulate repositioning of the breast from the prone to the supine positions, to help register regions of interest between diagnostic prone MRI and preoperative supine MRI. This involves creating a breast model from the prone diagnostic MRI, which is then embedded into the model. The supine position can be simulated by first determining the stress-free reference state of the breast using the methods described in Section 3.1, and then realigning the model with respect to the direction of gravity loading. The deformation field from this prone-to-supine repositioning simulation can then be used to warp the prone MRI to create a model generated supine MRI, which can then be more easily registered with the preoperative supine MRI using standard intensity-based nonrigid registration techniques.

Fig. 6B. Different approaches have been proposed to register the prone and supine MR images, including fluid registration (Carter et al., 2008) and B-Spline free-form deformation (FFD) registration (Lee et al., 2010a; Eiben et al., 2013; Han et al., 2014).

Biomechanical models have also been developed to determine the locations of tumors in the breast during surgery using stereoscopic camera systems (Carter et al., 2005; Carter et al., 2006; Carter et al., 2008). Carter et al. (2008) presented the first example of how such models could be used during surgery. In this approach, dynamic contrast-enhanced MR (DCE-MR) images were acquired in the prone position to identify the location of the tumor prior to surgery. Noncontrast-enhanced MR images in the supine position were also acquired prior to surgery, and used to construct a patient-specific model of the breast. The deformation from the supine to prone position was then simulated, allowing the tumor location identified in the prone DCE-MR images to be mapped to the supine model.

One confounding factor with this approach is that, during surgery, the supine position of the patient on the operating table may be different to their position in the previously acquired supine MR images, which were used to create the supine model. To account for this potential deformation and find the location of the tumor during surgery, a stereoscopic scan of the breast surface is obtained while the patient is on the operating table, and this is used to provide boundary conditions for deforming the supine model to the surgical configuration. The main motivation behind this approach is the relatively small difference in breast shape from when the patient was supine in the MR scanner to when the patient was lying the surgical table. This is used to justify the use of small-strain linear elasticity theory to predict the location of the tumor during surgery in a short time (45 s). Detailed reviews of the different applications of soft tissue modeling in image-guided surgery are provided by Carter et al. (2005) and (Hawkes et al., 2005).

5.3 PREDICTING OUTCOMES OF BREAST AUGMENTATION PROCEDURES

Biomechanical modeling of the breast can be used to predict surgical outcomes of BCT procedures, which involve the localized removal of tumors (del Palomar et al., 2008). In cases where BCT is not feasible, mastectomy is usually performed, which involves removal of all or most of a patient's breast. In such situations, women often choose breast reconstruction following surgery. Furthermore, there is a growing demand for breast augmentation for nonclinical, cosmetic or postural reasons, such as in the case of congenital asymmetry of the breasts (Roose et al., 2005), or for breast reduction. Many different implants are available to provide symmetrically shaped breasts (Spear and Spittler, 2001). It is therefore difficult to ascertain preoperatively which implant would be best for a given individual (Hudson, 2004). To address this issue, biomechanical models of the breast have been created to help predict surgical outcomes following such augmenting procedures (Roose et al., 2005; Roose et al., 2006a; Roose et al., 2006b).

6 TRANSLATING BIOMECHANICAL MODELS

In order for biomechanical models of the breast to be successfully implemented in a clinical setting, a number of practical challenges need to be addressed. These are mainly related to how research-based workflows (where execution speed and ease of use are not essential) are translated into clinical workflows (where reliability, practicality, and fast processing times are very important) (Rajagopal et al., 2010).

One of the main challenges in applying breast biomechanical models in a clinical environment is the availability of suitable images from which patient-specific anatomical models can be generated. Presently, MR images of the breast are routinely used in research applications to construct biomechanical models. However, acquisition of such images is expensive and time consuming. Furthermore, X-ray mammograms are usually the starting point for breast cancer diagnosis. This presents a significant challenge, because information pertaining to the geometry and 3D structural arrangement of breast tissues is unavailable from clinical screening programs. Studies are being performed to address this issue by approximating the 3D shape of the uncompressed breast from the combination of 2D mammograms, and a previously defined generic breast model (Qiu, 2009). In more recent studies, 3D ultrasound computer tomography (USCT), an emerging imaging modality that has been developed to overcome the drawbacks of X-ray mammography (Gemmeke and Ruiter, 2007), has been investigated for constructing biomechanical models of the breast (Hopp et al., 2015). During USCT imaging, the breast is suspended in a water bath. This imaging modality therefore also has the benefit of providing an estimate of the stress-free reference state of the breast. However, the clinical benefits of this imaging modality are still being investigated.

It is well-established that the predictions of breast biomechanical models are sensitive to the constitutive parameters used to describe the behavior of the tissues. For example, Hsu et al. (2011) performed an analysis of FE model parameters used for compression of a 3D breast phantom. They found that small alterations in the stiffness of the different tissue properties of the breast, or the friction between the skin and the compression plates, can change the displacements of the compressed tissues by more than 10 mm. This may have an impact when registering prone MR images with CC and MLO mammograms. Robust identification of the constitutive parameters is thus necessary in order to obtain precise predictions of tissue motion. As described in Section 5, current methods for identifying these parameters require additional clinical images, such as supine MR images, or MR images of the compressed breast in the prone position (MR-mammography images). However, these images are not routinely acquired in clinical practice. To address this issue, alternative methods for identifying the constitutive parameters of breast models have been investigated. For example, the use of a hand-held ultrasound indentation system was described by Han et al. (2002) and Goenezen et al. (2012) for in vivo identification of local breast tissue properties. Babarenda Gamage et al. (2011) investigated the use of multiple gravity loaded orientations for improving the identifiability of constitutive parameters. Such an approach may provide a cost-effective method for identifying patient-specific constitutive parameters, especially since the feasibility of scanning the surface geometry of the breast surface has been demonstrated using a range of commercially available scanning systems, including devices based on structured light reconstruction (Catherwood et al., 2014; Henseler et al., 2014; Wheat et al., 2014), laser scanning (Kovacs et al., 2007; Tepper et al., 2008; Eder et al., 2013), or stereoscopy (Lee et al., 2011; Donfrancesco et al., 2013; Khatam et al., 2015; Reece et al., 2015). However, these methods have not yet been demonstrated in practice for identifying constitutive parameters of patient-specific biomechanical models.

7 CONCLUSIONS

This review described the state-of-the-art in the field of modeling breast biomechanics, and highlights methods that have been used for constructing biomechanical models of the breast, and the constitutive relations that are typically used to describe the mechanical response of the different tissues. Examples were presented of how models have been used to address clinical problems related to colocating tumors in different diagnostic imaging modalities, and to track tumors for surgical applications. While substantial progress has been made in this field in recent years, a number of technical and practical challenges remain. Current technical challenges include the development of cost-effective methods for creating patient-specific biomechanical models of the breast, and the characterization of individual-specific mechanical properties of the different breast tissues. Practical challenges include translating research-based breast imaging and modeling workflows into clinical practice, and providing clinicians with easy to use graphical interfaces for interpreting the new information provided by the biomechanical models. Once these challenges have been addressed, there is great potential for breast biomechanics to improve outcomes for breast cancer patients.

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